

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

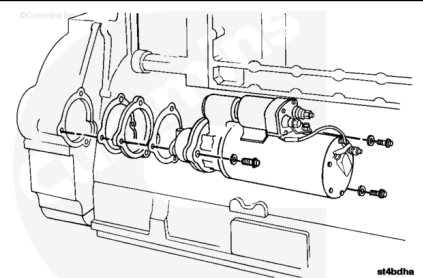
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Disconnect the electrical connections from the starting motor.



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.



Remove the starting motor capscrews, starter, spacers, and gaskets.

Discard the gaskets.

Note : All engines do not contain spacers and gaskets.

Clean and Inspect for Reuse

⚠ WARNING ⚠

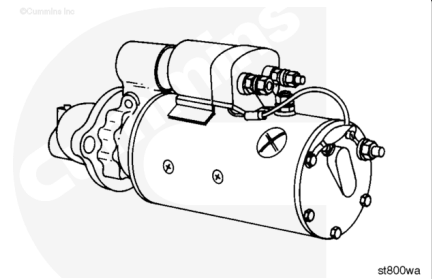
When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the exterior of the motor with steam.

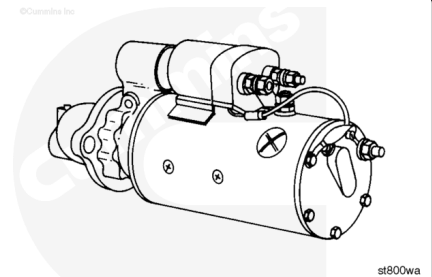
Use compressed air to dry the starter.



Inspect the gear, shaft, and the bushing for wear or damage.

Lubricate the bushing with engine oil.

Note : The pipe plug must be removed to lubricate the bushing on some starter motors.



Install

⚠ WARNING ⚠

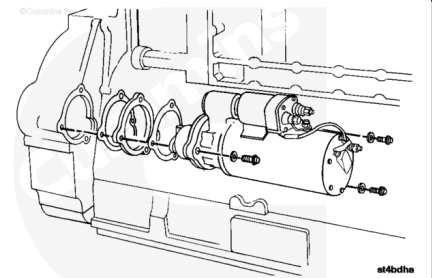
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Note : Not all engines use spacers.

Note : The wet type flywheel housing requires gaskets for the starting motor.

Install any spacer or gaskets.

Install the starting motor and capscrews.



Use the following torque values to tighten the capscrews.

Torque Value:

Cast Iron Flywheel Housing

1. 215 n•m [159 ft-lb]

Torque Value:

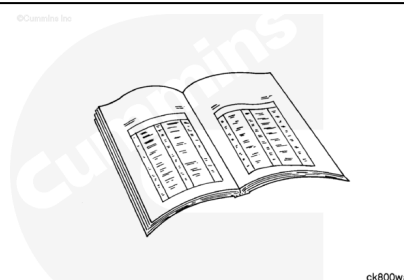
Aluminum Flywheel Housing

1. 195 n•m [144 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the electrical connections to the starting motor.
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.